Cognitive and Athletic Performance Optimization Using Machine Learning and Biometrics

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***Abstract*—This project investigates the relationship between biometric data, nutritional habits, and cognitive performance. The main goal is to build Machine Learning models able to predict the C-score, a measure of cognitive performance divided into three different components: memory, reasoning, and verbal ability.**

The data used in this work come from two datasets collected during the same experimental study. During the observation period, each participant recorded their meals, while wearable devices collected biometric information such as sleep data, heart rate, and respiration rate. In addition, participants performed cognitive tests from which the C-score was obtained. The nutritional dataset contains meal-level information associated with each cognitive test session, while the wearable dataset contains physiological and cognitive data related to the same subjects and testing dates.

The prediction task was formulated as a classification problem. For each cognitive target, the original continuous scores were divided into three classes, corresponding to low, medium, and high performance. The class thresholds were defined using the quantiles of each target distribution, producing approximately balanced classes. To reduce the risk of overly optimistic results caused by multiple observations from the same subject, the models were evaluated using Leave-One-Subject-Out cross-validation.

A dedicated model selection pipeline was developed to automate feature selection, hyperparameter tuning, and model comparison. For each target, several models were tested, including Logistic Regression, Random Forest, XGBoost, LightGBM, and Support Vector Classifier. After selecting the best models, an explainability pipeline based on SHAP, ALE, and DiCE was applied to interpret the contribution of the most relevant variables and identify possible relationships between biometric, nutritional, and cognitive factors.

The results suggest that variables related to sleep and physiological variability are often associated with cognitive performance, while some nutritional aspects also appear as possible factors worth further investigation. However, due to the limited size of the dataset, these findings should be interpreted as preliminary indications rather than causal conclusions.

I. Introduction

Cognitive performance is influenced by several physiological and behavioral factors, such as sleep quality, nutritional habits, heart rate, respiration rate, and stress. In recent years, wearable devices have made it possible to collect many of these signals in a continuous and non-invasive way. This creates the opportunity to study how daily biometric and lifestyle variables are related to cognitive outcomes, and whether Machine Learning models can use this information to predict different aspects of cognitive performance.

The **objective of this project** is to develop a complete data science pipeline for the prediction and interpretation of cognitive performance in relation to nutrition and wearable data. In particular, the work focuses on the C-score, which is divided into three cognitive components: **memory, reasoning, and verbal ability**. For each of these targets, the original continuous score was transformed into a three-class prediction problem, distinguishing between low, medium, and high performance.

The input data come from two sources collected during the same experimental study. The first dataset contains wearable-derived biometric information and cognitive test results, while the second contains nutritional data related to the meals consumed before each cognitive test session. After data cleaning and integration, these sources were combined into a single dataset including sleep-related variables, heart rate, respiration rate, demographic information, nutritional features, and cognitive targets.

The proposed workflow is organized into two main parts. The first one is the **Machine Learning pipeline**, which includes dataset integration, preprocessing, feature engineering, subject-aware scaling, feature selection, hyperparameter optimization, and model evaluation. Since multiple observations are available for the same subject, the models are evaluated using Leave-One-Subject-Out cross-validation, in order to reduce subject-level data leakage and obtain a more realistic estimate of generalization performance.

The second part is the **Explainability pipeline**, which is applied after model selection. Its goal is not only to evaluate predictive performance, but also to understand which variables are most relevant for the models and how they influence the predicted cognitive class. For this reason, SHAP is used to analyze feature importance, ALE is used to study feature effects, and DiCE is used to generate counterfactual examples when supported by the selected model.

The general architecture of the system, including the connection between the Machine Learning and Explainability pipelines, is shown in Figure 1.

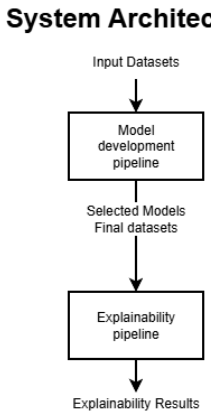


Fig. 1: Graphical representation of the general system architecture and the flow of data within it.

The rest of the document is organized as follows. Section II describes the Machine Learning pipeline, including dataset integration, preprocessing, feature engineering, validation strategy, feature selection, and model optimization. Section III presents the Explainability pipeline and explains how SHAP, ALE, and DiCE were used to interpret the selected models. Section IV reports and discusses the main results for memory, reasoning, and verbal ability. Finally, the last section summarizes the main conclusions, limitations of the work, and possible future improvements.

II. Machine Learning Pipeline

This section describes the Machine Learning pipeline developed to transform the raw biometric, nutritional, sleep-related, and demographic data into models able to predict the three cognitive targets. The pipeline starts from the integration of the available datasets and ends with the selection of the best model configuration for each target.

The main stages of the pipeline are:

1) Problem Understanding and Dataset Integration
2) Exploratory Data Analysis and Data Cleaning
3) Feature Scaling and Target Discretization
4) Feature Engineering and Leave One Subject Out Cross Validation
5) Iterative Feature Selection and Hyperparameter Optimization
6) Best Model Selection and Performance Evaluation
7) Data Exporting

A. *Problem Understanding and Dataset Integration*

The task was predicting cognitive performance from user data, including biometric, nutritional, sleep-related, and demographic information. Therefore, the problem is formulated as a **supervised learning task**, where the input features describe the physiological and behavioral state of a subject around a given testing day, and the output is the corresponding **cognitive performance score**.

Two datasets were used: the first one contains wearable-derived biometric data and cognitive test results. Each record is associated with a subject and a specific test date. The wearable data include sleep-related variables, stress information, heart rate (HR), respiration rate (RR), and cognitive scores.

TABLE I: Input Datasets and Integrated Dataset Structure

Dataset	Granularity	Main Variables	Role
Wearable and cognitive dataset	Subject–testing session	HR, RR, sleep, stress, C-score	Biometric input and targets
Nutritional dataset	Subject–meal–temporal window	meal type, proteins, fats, carbohydrates	Meal-level nutritional input

Feature Group	Examples	Description
Identifiers	userID, reference date	Keys used for alignment and LOSO grouping
Demographic data	sex, age, height, weight	Static subject-level variables
Sleep data	sleep time, deep sleep, REM sleep	Recovery information around the test session
Wearable signals	HR day/night, RR day/night, stress	Raw physiological time-series data
Nutritional data	meal type, proteins, fats, carbohydrates	Meal-level intake in the temporal window around the test
Cognitive targets	memory, reasoning, verbal	Components of the cognitive score

The second dataset contains nutritional information collected around the same cognitive testing sessions. Each record includes meal-level **nutritional variables** like macros of the meal and a **categorical meal identifier**, where different values represent breakfast, morning snack, lunch, afternoon snack, and dinner. Since nutrition may influence cognitive performance depending both on daily intake and meal-specific composition, these variables were preserved during the integration phase.

Before training the models, the two datasets were merged using the subject identifier and the test date as common keys. The resulting dataset combines biometric, sleep, nutritional, demographic, and cognitive information into a unified dataset.

The two datasets have different granularities: wearable and cognitive data are organized at subject–testing-session level, while nutritional data are organized at subject–meal level within a temporal window surrounding the cognitive test day. Therefore, before merging, the nutritional dataset required the extraction of both the subject identifier and the reference date from the original compound ID field which followed the format ID_USERID_DATE. Then, the two sources were aligned using the subject ID and the reference testing date as common keys.

Table I summarizes the input datasets and the resulting integrated representation.

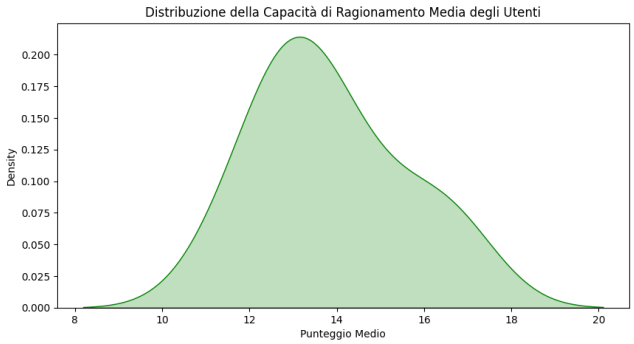
B. *Exploratory Data Analysis and Data Cleaning*

The final merged dataset was inspected before model training in order to understand its size, missing-value structure, and target distributions. After dataset integration and feature extraction, the dataset contained 150 observations and 140 columns. After removing unusable subjects, dropping fully uninformative columns, filtering invalid target values, and completing the cleaning procedure, the final dataset used for modelling contained 84 observations and 135 columns. After separating the targets and removing non-numerical or identifier columns, the final feature matrix contained 123 input features after the feature engineering phase described later.

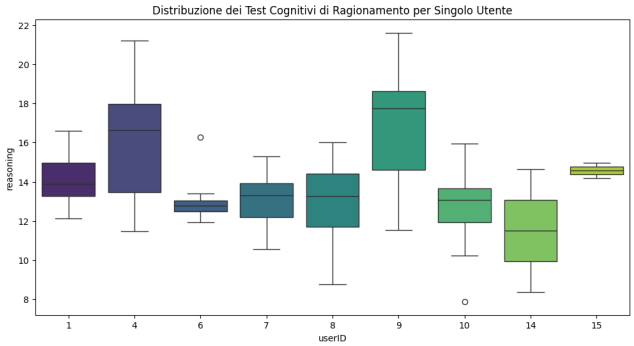
The **missing-value analysis** showed that most missing values were concentrated in wearable-derived HR and RR features. Before the final imputation step, some of these variables contained 29 missing values, corresponding to about 26.6% of the available observations. After filtering observations with valid cognitive targets, the amount

of missing data was strongly reduced, with the most affected features containing only 4 missing values, corresponding to about 4.76%. To preserve information about the original missingness pattern, four binary indicators were added: `HR_day_missing`, `HR_night_missing`, `RR_day_missing`, and `RR_night_missing`.

An exploratory data analysis was also performed to inspect the distribution of cognitive targets and biometric variables across subjects. In particular, the distribution of memory, reasoning, and verbal scores was analyzed both globally and at subject level. This step highlighted a relevant inter-subject variability: different users showed different baseline cognitive scores and physiological patterns.



(a) Distribution of the average reasoning score across subjects.



(b) Subject-level distribution of reasoning test scores.

Fig. 2: Exploratory analysis of the **reasoning target**. The first plot shows the distribution of subject-level average scores, while the second plot highlights variability across repeated test sessions for each subject.

This aspect is important because the objective of the model is not only to distinguish subjects with different average abilities, but to learn how physiological, nutritional, and behavioral variations around a testing session are associated with changes in cognitive performance.

The main preprocessing and cleaning operations applied before model training are summarized in Table II.

TABLE II: Main Preprocessing Operations

Step	Operation	Purpose
Column standardization	Rename shared variables	Align dataset schemas
Categorical encoding	Map categorical values to numbers	Enable ML processing
Date extraction	Parse date from nutrition ID	Create merge key
Subject extraction	Extract integer user ID	Create subject key
Dataset merge	Join on userID and date	Build unified dataset
Column removal	Drop redundant or empty columns	Reduce noise
Target filtering	Remove rows without cognitive scores	Enable supervised learning
Missing-value handling	Subject-aware imputation	Preserve individual baselines
Missingness indicators	Add imputation flags	Retain missing-data information

C. *Feature Scaling and Target Discretization*

The EDA highlighted strong inter-subject variability. For this reason, a **subject-wise scaling** strategy was adopted for dynamic variables. This transformation reduces inter-subject baseline differences and allows the model to focus on within-subject deviations.

Given a feature x and a subject s, each value was scaled using statistics computed only from the observations of the same subject:

$$x'_{i,s} = \frac{x_{i,s} - \text{median}(x_s)}{\text{IQR}(x_s)} \tag{1}$$

where $x_{i,s}$ is the original value of observation i for subject s, $\text{median}(x_s)$ is the median value of that feature for subject s, and $\text{IQR}(x_s)$ is the corresponding interquartile range, so the distance between the 25th and 75th percentiles, measuring the **spread of the middle 50%** of data. A robust scaling strategy was preferred because it is **less sensitive to outliers** than standard z-score normalization.

Static features, such as sex, age, height, and weight, were not scaled at subject level because they are constant for each subject. Applying subject-wise scaling to these variables would collapse them to zero and remove their informative content. For this reason, they were scaled globally using a StandardScaler when required by the model.

Although the original cognitive scores are continuous variables, the prediction task was also formulated as a classification problem. Each cognitive target, namely memory, reasoning, and verbal ability, was discretized into three performance classes: low, medium, and high.

Instead of using fixed thresholds, the class boundaries were defined using empirical quantiles. Specifically, the 33rd and 66th percentiles of each target distribution were used as thresholds. This choice allowed the classes to be almost equally represented, reducing class imbalance and making the classification task more stable.

Given a cognitive score y, the class label c was assigned as follows:

$$c = \begin{cases} 0, & y < q_{0.33} \\ 1, & q_{0.33} \leq y \leq q_{0.66} \\ 2, & y > q_{0.66} \end{cases} \tag{2}$$

where $q_{0.33}$ and $q_{0.66}$ are the empirical quantiles computed on the corresponding cognitive target. Class 0 represents low performance, class 1 represents medium performance, and class 2 represents high performance.

TABLE III: Quantile-Based Discretization of Cognitive Targets

Target	$q_{0.33}$	$q_{0.66}$	Low	Medium	High
Memory	12.66	14.99	28	27	29
Verbal	12.43	14.12	27	28	29
Reasoning	13.04	14.93	28	27	29

As shown in Table III, the quantile-based discretization produced almost balanced classes for all three targets. This is particularly important because it avoids training models biased toward a dominant class and allows accuracy-based evaluation to be more meaningful.

D. Feature Engineering and Leave One Subject Out Cross Validation

This phase included the conversion of raw wearable time-series into fixed-length descriptors, the removal of non-numerical or non-informative columns, and the definition of a subject-independent validation strategy.

*1) **Time-Series Feature Extraction**:* The wearable dataset contained raw time-series signals for heart rate and respiration rate, collected during both daytime and nighttime windows. Since the models considered in this work require fixed-length tabular inputs, each raw signal was transformed into a set of statistical descriptors like means, standard deviation, skewness,...

Central tendency and dispersion were described through mean, standard deviation, median, minimum, maximum, and range. **Distributional properties** were captured using skewness, kurtosis, quartiles, and interquartile range. **Energy-related** descriptors included root mean square, signal energy, and absolute mean. **Temporal variability** was represented through first-order differences, such as mean absolute difference, maximum absolute difference, and standard deviation of differences. Finally, peak-based descriptors were extracted to capture **local fluctuations** in the physiological signal. This process converted each raw time series into a fixed-size feature vector, in which each row corresponds to one subject observation and each extracted descriptor is stored as a separate numerical feature.

TABLE IV: Extracted Features from Wearable Time Series

Feature Family	Extracted Features	Captured Information
Central tendency	mean, median	Average signal level
Dispersion	std, min, max, range	Signal variability
Distribution shape	skewness, kurtosis, q25, q75, IQR	Signal distribution profile
Energy	RMS, energy, absolute mean	Overall signal intensity
Temporal variation	mean/max absolute diff., std diff.	Short-term signal changes
Peak behavior	number of peaks, peak mean	Local physiological fluctuations

*2) **Leave-One-Subject-Out Cross-Validation**:* Leave-One-Subject-Out (LOSO) cross-validation was used to assess whether the models could generalize to new subjects whose data were not included during training.

At each fold, all observations belonging to one subject were held out as the test set, while the observations of all remaining subjects were used for training. This validation strategy is particularly important in the present setting because multiple observations may belong to the same subject. A random train-test split could place different observations of the same subject in both training and test sets, leading to subject-level data leakage and in that case the model could learn subject-specific baselines instead of generalizable relationships between variables.

Algorithm 1 Leave-One-Subject-Out Cross-Validation

Require: Dataset D, subject identifiers S, model M
Ensure: Cross-validation scores
1: **for** each subject $s \in S$ **do**
2: $D_{\text{test}} \leftarrow$ all observations of subject s
3: $D_{\text{train}} \leftarrow$ all observations of subjects different from s
4: Train M on D_{train}
5: Evaluate M on D_{test}
6: Store the fold score
7: **end for**
8: **return** Mean score across all folds

E. Iterative Feature Selection and Hyperparameter Optimization

After defining the validation strategy, the final model selection phase was designed as an iterative search over three main elements: the cognitive target, the feature selection strategy, and the candidate model. The objective was not only to compare different model families, but also to identify, for each target, the subset of features and the hyperparameter configuration that achieved the highest LOSO validation accuracy.

For each cognitive target, namely memory, reasoning, and verbal ability, three feature selection strategies were tested: forward selection, backward selection, and stepwise selection. Each strategy was combined with five candidate classification models: XGBoost, LightGBM, Random Forest, Logistic Regression, and Support Vector Classifier. Overall, the **pipeline evaluated 45 configurations**.

The **candidate models** were stored in a structured list containing the model name, the estimator class, the Optuna hyperparameter search space, and possible fixed parameters, such as the random seed. The models considered in this phase are reported in Table V.

TABLE V: Candidate Classification Models

Model	Estimator	Main Tuned Parameters
XGBoost	XGBClassifier	estimators, depth, learning rate, regularization
LightGBM	LGBMClassifier	estimators, depth, learning rate, leaves
Random Forest	RandomForestClassifier	estimators, depth, split and leaf size
Logistic Regression	LogisticRegression	regularization strength C
Support Vector Machine	SVC	C, kernel, gamma

For each model, three feature selection strategies were tested independently:

- **Forward selection**, which starts from an empty feature set and progressively adds the feature that provides the largest improvement in validation score.
- **Backward selection**, which starts from the complete feature set and progressively removes features whose exclusion improves or preserves the validation score.

- **Stepwise selection**, which combines forward addition and backward removal, allowing the selected subset to be refined during the search.

For each target and feature selection method, all candidate models were evaluated. First, the feature selection procedure was applied to identify a reduced subset of informative variables. Then, the selected features were used inside the Optuna optimization loop to tune the model hyperparameters. The optimization objective was the mean LOSO validation accuracy.

The full **procedure** was **repeated** for the three cognitive **targets**: memory, reasoning, and verbal score. Therefore, for each target, the pipeline evaluated every combination of feature selection strategy and candidate model. The final output was a results table containing, for each configuration and model, the selected features, their number, the best hyperparameters, and the corresponding validation score of the tested models.

Algorithm 2 Iterative Model Selection Pipeline

Require: Dataset D, targets T, model array M, feature selection methods S, Optuna trials N
Ensure: Results table with selected features, best hyperparameters, and validation score for each configuration

```
Initialize results table R_t
for each target t ∈ T do
    Select the target vector y_t
    for each feature selection method s ∈ S do
        for each model entry m ∈ M do
            Load estimator class and model-specific search space
            Initialize base model using static parameters
            Apply FS method s using model m and target y_t
            Obtain the selected feature subset F_{t,s,m}
            Build reduced feature matrix X_{t,s,m} using F_{t,s,m}
            Define the Optuna objective using X_{t,s,m}, y_t, and the model-specific search space
            Run Optuna optimization for N trials
            Store selected features, best hyperparameters, and validation score in R_t
        end for
    end for
end for
return Results table R
```

In this way, the final models are the result of the joint optimization of the feature selection strategy, the model family, the selected feature subset, and the corresponding hyperparameters.

F. Best Model Selection and Performance Evaluation

At the end of the iterative optimization procedure, the best configuration for each cognitive target was selected according to the highest LOSO validation accuracy. Table VI reports the best-performing model, the feature selection strategy, the number of selected features, and the final validation accuracy.

TABLE VI: Best Configuration for Each Cognitive Target

Target	Best Model	FS Method	# Features	Accuracy(LOSO)
Memory	LightGBM	Forward	2	0.639
Reasoning	SVC	Forward	8	0.718
Verbal	SVC	Stepwise	6	0.664

The results show that the best-performing model depends on the cognitive target. Memory was best predicted by a compact LightGBM model using only two daytime heart-rate descriptors, while reasoning and verbal ability achieved the highest scores with Support Vector Classifiers. This suggests that the relationship between biometric, nutritional, and sleep-related variables and cognitive performance is not uniform across cognitive domains. For this reason, a **separate model** was optimized for **each target** instead of using a single model for all cognitive components.

The **selected feature subsets** were also different across the three targets. For memory, the best LightGBM model selected only two heart-rate features: `HR_day_1h_rms` and `HR_day_1h_median`. For reasoning, the best SVC model selected a more heterogeneous subset including respiration, heart-rate, nutritional, and anthropometric variables. For verbal ability, the best SVC model mainly selected nighttime respiration, sleep, and heart-rate features.

TABLE VII: Selected Features of the Best Models

Target	Model	Selected Features
Memory	LightGBM	`HR_day_1h_rms`, `HR_day_1h_median`
Reasoning	SVC	`RR_day_1h_q75`, `Kcal_3`, `HR_day_1h_iqr`, `Carboidrati_3`, `RR_day_1h_std_diff`, `RR_day_1h_max`, `Grassi_5`, `weight_kg_x`
Verbal	SVC	`RR_night_imputed_mean`, `sleepTimeSeconds`, `carbs_over_proteins_4`, `RR_night_imputed_std_diff`, `HR_night_imputed_mean`, `RR_night_imputed_q75`

G. Data exporting

After careful consideration and analyzing the top models, we converged on the decision of passing 2 models rather than one onto the Explainability pipeline. As we believed it could help us have a more complete perspective on; the features' effects on the given cognitive scores. We did so while still keeping the distinction between the top (primary) model and the secondary model for the given target. The 3 extra selected models are those shown in table VIII

TABLE VIII: Second Best Configuration for Each Cognitive Target

Target	Best Model	FS Method	# Features	Accuracy
Memory	SVC	Forward	6	0.634
Reasoning	Logistic Regression	Backward	112	0.718
Verbal	Logistic Regression	Backward	114	0.664

To permit simple iteration the data is passed onto the explainability pipeline via dictionary type data structure that contains

- **A two-level dictionary** where the first-level keys are the 3 cognitive targets, the second-level keys are the model ranks (`primary`, `secondary`), and the data stored inside is the one that from the **results table**.
- **The final scaled dataset** without the targets,
- **A dictionary** containing all the target data frames utilized for model training

This dictionary will be called Export_ Bundle for when discussed in the next section for simplicity.

III. Explainability pipeline

This section covers the functioning of the Explainability pipeline. that is organized as shown in figure 3 with a non linear flow of information, utilizing the SHAP analysis as a basis for determining what features to perform the subsequent ALE analysis, and generate the DiCe counterfactuals on.

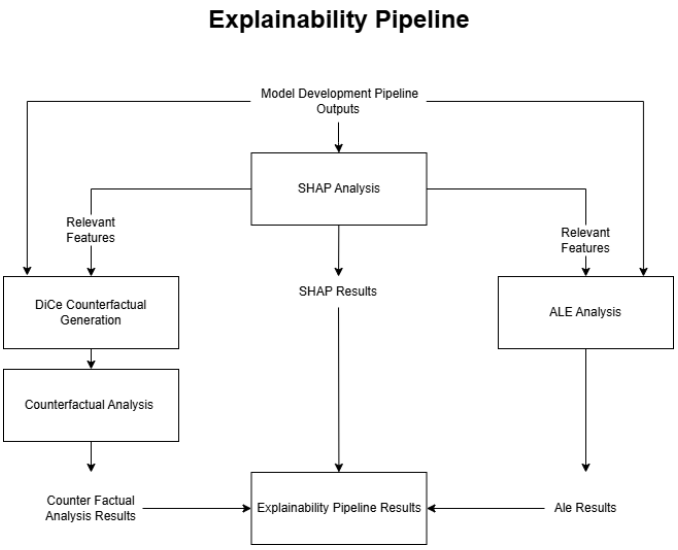


Fig. 3: Graphical representation of the flow of data within the Explainability Pipeline

All the analysis generated their data by following the process detailed by algorithm 3. That iterates through all the Models present in the export bundle's two level dictionary applying the algorithm relevant to the analysis performed as it goes.

The only major difference between what is shown by algorithm 3's pseudo code and what was actually applied is that the two analysis besides SHAP also require the data generated by the SHAP analysis to function.

1) SHAP Analysis: The SHAP analysis stage is responsible for generating feature importance information for the selected models obtained from the model development pipeline.

Using the processed dataset, the selected features, and the trained models, **SHAP values** are calculated for each Model based on its associated prediction target. These values quantify the contribution of each feature to the model predictions like that providing a measure of **feature importance.**

Algorithm 3 Interpretability Data Generation Pipeline

Require: Export Bundle E,
Ensure: Easily accessible Human interpretable data on feature importance, effects, and how their variation influences predicted performance

```
Create Data storage folder for the Analysis D_a
for Cognitive Area c and ModelRank dictionary m_r items in E do
    for Ranking r and Model Data m_d in m_r do
        Create data storage folder m_f with name c+r in D_a
        Initialize Data required for the specific analysis type
        Run analysis specific algorithm
        Store generated data in folder m_f
    end for
end for
```

The generated SHAP values are then used in two manners.

- To generate graphical representations of the feature contributions, allowing the most influential variables to be identified.
- To identify the most relevant features for subsequent explainability procedures.

There are 3 types of graphs that were generated during the SHAP Analysis phase:

- **A Bar Plot** that visually represents the mean SHAP value of the most influential features, used to visualize the importance of the features that will then be passed onto the other two analysis.
- **A Beeswarm Plot** That combines feature importance and feature effect information into a single visualization. Each point represents one observation in the dataset, while its position on the horizontal axis corresponds to the SHAP value associated with a particular feature.
- **A Waterfall Plot** That illustrates how individual feature contributions combine to produce a specific prediction. Starting from the model's baseline prediction, each feature either increases or decreases the predicted outcome according to its associated SHAP value.

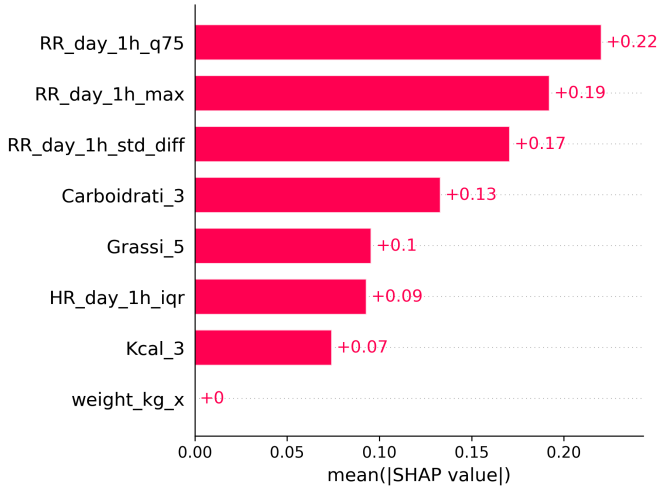
Examples of each of these graphs are shown in figure 4.

As can be seen in these examples the SHAP analysis permits us to notice things that we normally would be unable to without it. Such as the fact that weight has no effect on the models prediction in spite of it having been selected by the model. This thus permits us to discard it before starting the subsequent analysis phase.

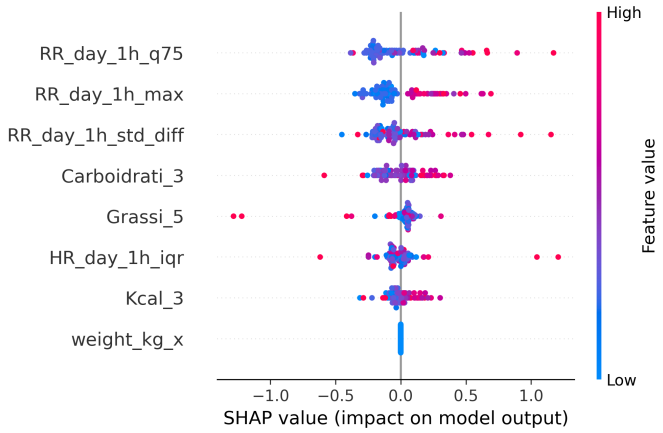
A. ALE Analysis

The ALE analysis stage consists of analyzing the influence of the 7 features with the highest `mean(|SHAP Value|)` of each of the models. Through the generation of a Accumulated Local Effects (ALE) plot for each.

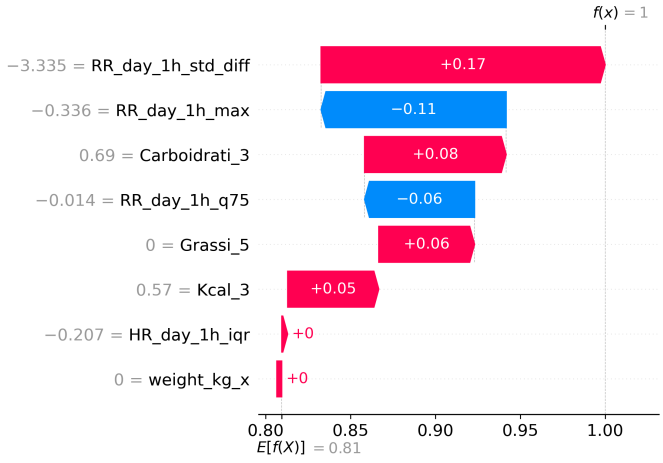
These ALE plots provide a graphical representation of how variations in a given feature affect the predicted outcome while also accounting for the distribution of the real data. Figure 5 is an example of one of the ALE plots generated



(a) Bar Plot Graph



(b) Beeswarm Graph



(c) Waterfall Graph

Fig. 4: Graphics generated during the SHAP Phase for the Forward Selection SVC model trained on the Reasoning Dataset

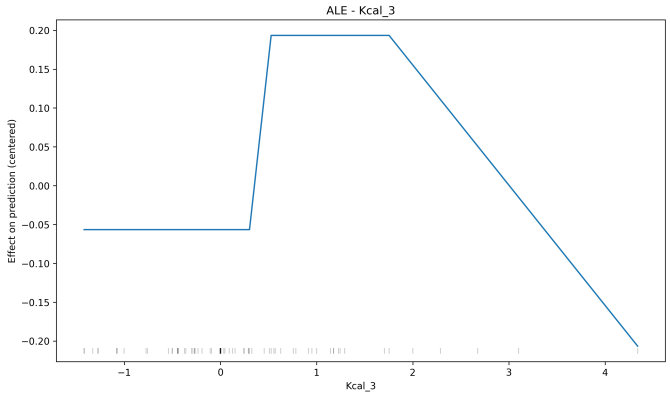


Fig. 5: ALE plot for effect of Kcal on meal 3 on reasoning performance, as described by the SVC model that selected its features via forward selection and was trained on the reasoning dataset

To read the ALE graph it is very important to pay attention to the ticks visible above the x axis as they represent the feature distribution in the dataset. For example from this example we can confidently infer that there is a certain caloric intake minimum necessary for reasoning performance maximization.We can also maybe conclude that there is a point where continued increase in caloric intake causes a decrease in performance. But this is less certain as there is less dataset specific data to corroborate this.

B. DiCe Counterfactual generation

The DiCe Counterfactual generation stage consists of generating a series of counterfactuals that only modify the 5 most relevant features identified during the SHAP analysis stage.

The counterfactuals are generated by selecting 7 random results from the final dataset then making the DiCe algorithm generate 5 counterfactuals for each of the other 2 possible target results. These are saved in a results table that also contains a column detailing which features where modified by the DiCe algorithm and by how much.

A major constraint encountered with this phase is that the SVC models selected were trained without a probability function,thus making the counterfactual generation for these impossible.It is because of this that the DiCe phase could only be performed, on the non SVC of models.

C. Counterfactual Analysis

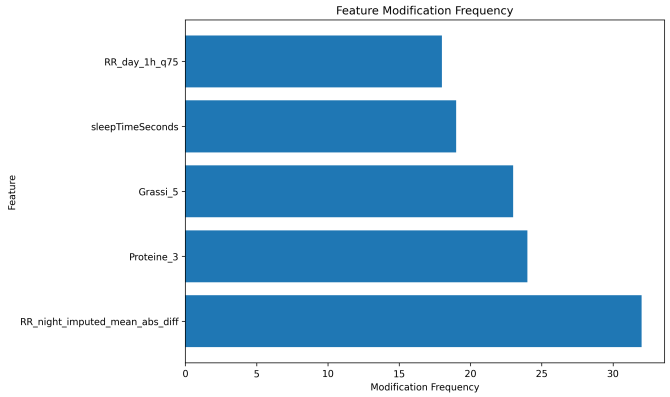
Since the data generated by the previous phase is uninterpretable by humans, counterfactual analysis phase consists in turning the data generated during the previous phase into a human interpretable version.

It does so by generating 2 data frames that are formatted as goes.

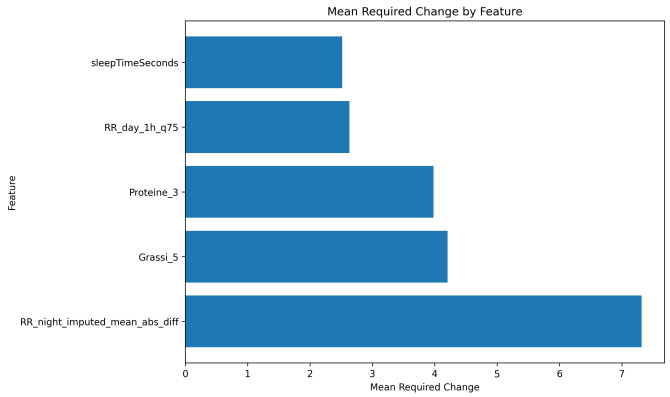
- One that shows date change frequency of the features,
- One that shows the Mean Required Change in said features to achieve the desired result.

Alongside the tables bars graphs were also generated. With the objective of making the analysis of the data as simple as

possible. Figure 6 shows examples of the graphs generated for two of the tables.



(a) Feature Modification Frequency Graph



(b) Mean Required change by feature bar graph

Fig. 6: Example of graphs generated during the counterfactual analysis phase belonging to the Logistic Regression whose features were selected using backward feature selection and was trained to predict reasoning scores

Due to the aforementioned constraint that hampered our capability of generating DiCe counterfactuals for all the models. We decided on making only the simple graphs shown in figure that take the secondary role of helping us corroborate or disprove the findings we gathered from the SHAP and ALE analysis. For example with the graphs visible here **RR_day_1h_q75** had the highest `mean(|SHAP|)` score. Yet it was the least modified one by DiCe and had a flat ALE graph pointing us to the conclusion that this feature is likely **not as important as it seemed** at a first glance.

IV. Results

This section will cover the results after the thorough analysis of all the data produced by the explainability pipeline it will be divided by cognitive area, and have a final conclusions area at the end.

A. Memory

With regards to our analysis of the features on memory performance it proved to be the least conclusive by far

First we were forced to disregard the primary model (LightGBM) because first the SHAP and Dice were inconclusive.Second it selected only two features pointing to possible overfitting or finding a fake pattern in the small data pool,and lastly the ALE plots for both features do not exhibit a clear monotonic relationship with the predicted target. As while an increase in the feature appears to positively influence the prediction within a limited range, the effect becomes irregular for larger values. Given the limited number of observations available in some regions of the feature space, these fluctuations become hard to interpret.

all selected features point to A **high respiratory rate variability during the day being correlated with better performance** This is because the two features with the highest SHAP score are **RR _Day _1h _Mean _ABS _diff** and **RR _Day _1h _rms** which as their name implies they are related to variability during the day, and as visible in the figure 7 they both point to the same general conclusion according to SHAP,and more importantly the same pattern is also supported by the ALE plots of both features

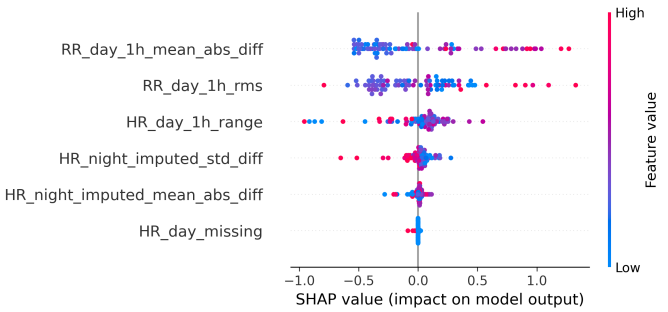


Fig. 7: Beeswarm graph from the SHAP analysis of the SVC whose features were selected with forward selection and was trained on the memory target

This alone might seem like a peculiar result until someone considers the possibility that High RR variability is likely tied to exercise or some form of exertion during the day. So although not definite one can conclude that there might be a relationship between exercise/exertion and memory performance,so looking further into this with a bigger data pool could prove useful.

B. Reasoning

The reasoning analysis on the other hand was definitely more conclusive. From our analysis we could conclude that:

- That there is a likely a **positive correlation** between **sleep time** and **reasoning performance** as concluded from the 3 analyses on the secondary model which was the one that contained the feature, and had both it and **awake sleep time** as top `mean(|SHAP|)` score features, and pointing towards the same conclusion
- That there is a possible negative effect of **high fat intake during meal 5** on the score as both models selected the feature agreed on it

- and the last conclusive data point pointing to a likely correlation between **RR variability** and **better cognitive performance**. Which as in the other case points to exercise, and was supported by ALE and SHAP analysis of reasoning's forward selection **SVC** that selected 3 features related to this all pointing toward the same conclusion in both analyses

C. Verbal

Verbal was perhaps the area where we got the most conclusive answers with both the primary and secondary models selecting multiple common features and having them as their top features according to `mean(|SHAP|)` scores.

We do end up arriving to similar conclusions as reasoning with the most clear conclusions being:

- Sleep again with **sleeptimeseconds** appearing in both and **awakeSleepSeconds** appearing in the secondary model again pointing to the same thing higher sleep is better
  - A possible correlation between **higher respiration rate variability at night** and lower performance maybe pointing at effects of apnea induced poor sleep but inconclusive without further information.
  - A positive correlation between a higher Heart Rate inter quartile range and verbal performance. Again pointing to better sleep
- A possible negative correlation for a higher carbs to protein ratio with **both models having selected this variable for different meals of the day** and pointing to the same conclusion
- Again a possible correlation between **higher fat intake at dinner** and lower performance.

V. Conclusion

A. Main Conclusions

Concluding, after the thorough analysis of the results a exposed during the previous section. It can be said that:

- The most clear take away, that is echoed by most of the models, and corroborated in the different types of explainability analysis is. That there is a clear correlation between good sleep quality, and cognitive performance the following day.
- Less confidently it can also be said that the models also point towards a positive correlation between higher exercise during the day and cognitive performance
- Interestingly two less clear correlations that point toward conclusions that are not as expected as the other 2 which are already well proven and documented factors
  - Firstly we identified that there is a possible negative correlation between higher fat intake during meal 5 and, reasoning and verbal performance
  - Secondly a possible negative impact of a higher carb to protein ratio on performance.

Though all of these conclusions should be taken with caution. Since due to the limitations that will be exposed during the next subsection. What was achieved during this project should only be taken as a guidance as to where to dedicate future efforts towards with the hopes of seeing whether the results are proven correct or just the results of patterns relative to the given dataset.

B. Limitations

As already briefly covered while discussing the different pipelines and results we encountered a series of limitations during the development of the project. Which although they didn't stop us from finalizing the project and getting the presented results. Should still be taking into consideration when drawing conclusions of what is exposed in the document, they are:

- The small size of the input data sets
  - the Data sets provided to us, for the project only had 150 obesrvation spread across 15 unique users on whom the tests were conducted. This with 140 columns most of which were feature data. Post data processing the situation only got worse as we were forced to eliminate a significant portion of the data rows due to either incomplete data or unusable data. This means that:
    1) All the model training was performed on a diminute dataset, that means that even with the utilization of advanced techniques like Leave One Subject Out the risk of overfitting the model is something that cannot be disregarded.
    2) Losing out on possible correlations between features and targets due to it being to weak for the pattern to be picked up by the models in such a small dataset
    3) As already mentioned during the previous subsection, it would be necessary to perform this analysis once more with a substantially larger data pool so that we can corroborate the results obtained.
- The selected SVC models not being trained with the Proba function set as true
  - This was more a training resources driven decision as during the model training process, as we discovered that setting proba_function to true greatly increased training times leading to session disconnects within the colab before the process could be finished
    1) The limitation the comes from the fact that SVCs configuration were some of the most performant models in all of the target areas. This meant that they were passed onto the explainability pipeline were one problem arises. DiCe counterfactual generation needs a model with a probability function to function. This as already briefly covered in prior sections meant that, the DiCe analysis could only be performed on half the selected models and had to take complementary role. As opposed to the primary role we had intended for it to have in the overall results analysis.

- Lack of experience
  - Although not a limitation from the project itself it should always be noted it was our first time performing this type of analysis, and there always is a chance of us making mistakes we are not aware of. Though of course this is also the point of the experience and could be also seen as an advantage since we were able to learn an important amount of things from embarking on this journey and completing this project.

C. What can be done in the future

Considering the types of limitations faced and the conclusions that were able to be drawn from the explainability results analysis, a series of things that can be done on the future to build on what has been achieved through this project are:

1) Perform a new study on a larger subject base that is also performed for a longer amount of time, making it more focused on one of the given areas so as to not waste resources making it overly general, which could limit analytical capability
   - The areas where we were able to find conclusions could be used as a basis on where to concentrate the study's efforts
2) Dedicate more resources into model training, to train the SVC models with Proba functions set as true
3) Utilize the experience gathered while developing this project to streamline the pipelines and automate the areas where manual inpu/selection was required.
   - To allow us to re run it on distinct data sets without having to manually modify code each time.
4) If more concrete results are achieved train a Large Language Model on them
   - With the goal of teaching it how to help users to improve their cognitive capabilities with simple modifications be them dietary and/or lifestyle based (like improving sleep quality and exercise)

D. Final Words

Closing off, while the scope of this project was limited by the size and duration of the study, we can confidently say the results demonstrate the potential of explainable machine learning techniques to identify meaningful relationships between lifestyle, dietary, and cognitive-performance-related factors. The findings obtained provide a foundation for future research involving larger and more focused datasets, improved model training methodologies, and greater pipeline automation. Ultimately, this work represents a step toward the development of intelligent systems capable of providing personalized, evidence-based recommendations that help individuals improve their cognitive capabilities through simple lifestyle and dietary modifications, and having time we would be glad to continue working towards this goal one small step at a time.